

1. 5 Core Components of Data Communication

- 1. **Message:** Actual data payload to communicate (Text, Number, Audio, Video).
- 2. **Sender:** Workstation/phone generating the source bit stream.
- 3. **Receiver:** Sink device accepting the delivered information.
- 4. **Transmission Medium:** Guided (Twisted pair, Coax, Fiber) or Unguided (RF).
- 5. **Protocol:** Agreed rules governing Syntax, Semantics, and Timing.

Effectiveness: Delivery (Right node), Accuracy, Timeliness, Jitter (Delay variance)

2. Data Flow & Transmission Modes

Mode	Direction	Channel Allocation	Real-World Example
Simplex	Strictly 1-Way (Unidirectional)	100% capacity in 1 direction	Keyboard to CPU, TV broadcast
Half-Duplex	2-Way, but 1 at a time	Alternating time sharing	Walkie-Talkie ("Over & Out")
Full-Duplex	Simultaneous 2-Way	Divided / Separate channels	Telephone Call, Switched Ethernet

Connections: Point-to-Point (Dedicated link) vs Multipoint (Shared capacity among >2 nodes).

3. Physical Network Topologies Formula & Trade-off Matrix (for n Nodes)

Topology	Physical Links Formula	I/O Ports/Node	Cabling Cost	Fault Tolerance Profile	Key Failure Bottleneck
Mesh	$n(n - 1) / 2$	$n - 1$	Highest $O(n^2)$	Maximum; Dedicated alternate routes	Excessive wiring & physical port saturation
Star	n	1	Moderate $O(n)$	High (1 broken cable isolates only that host)	Central Hub / Switch Failure collapses entire network
Bus	1 backbone + n drops	1 tap	Lowest $O(1)$	Poor (Backbone cut halts entire bus)	Cable break or missing terminator stops all reflections
Ring	n (Unidirectional)	2 (In / Out)	Low to Moderate	Single point of failure (Dual ring FDDI fixes)	Token circulation break halts transmission
Tree / Hybrid	Hierarchical ($\$n-1\$$)	1 per leaf	Moderate to High	Isolated to sub-branch	Root hub / Trunk line failure disconnects branches

4. Geographical Network Categories

Category	Range	Typical Medium	Standard
PAN	< 10 m	Bluetooth, Zigbee, IrDA	IEEE 802.15
LAN	100 m - 2 km	Twisted Pair, Wi-Fi	IEEE 802.3, 802.11
MAN	5 - 50 km	Optical Fiber, Coaxial	Cable TV, Metro Eth
WAN	> 100 km	Submarine Fiber, Satellite	IP, BGP, MPLS

5. 3-Tier ISP Architecture & Internet Structure

- Tier 1 ISPs:** Global backbone carriers (AT&T, Lumen, Tata). Own trans-oceanic cables. *Zero transit fees*; free settlement peering.
- Tier 2 ISPs:** Regional operators (Airtel, Jio). Buy transit from Tier 1; peer at **IXPs (Internet Exchange Points)**.
- Tier 3 ISPs:** Local retail broadband (ACT, Hathway, FTTH). Last-mile access to homes and businesses.
- POP (Point of Presence):** Physical interface facility where external networks hook into an ISP backbone.

6. Protocols, Standards Organizations & Layering Principles

Protocol 3 Elements:

- Syntax:** Data structure & bit field layout.
- Semantics:** Meaning & control actions.
- Timing:** Speed matching & sequencing.

De Jure vs De Facto:

- De Jure:* Legally approved by treaty bodies (ISO 7498, IEEE 802.3).
- De Facto:* Established by industry adoption without formal law (TCP/IP).

Standards Organizations:

- ISO:** OSI Reference Model (7 Layers).
- IEEE:** LAN standards (802.3, 802.11).
- IETF:** Internet Standards via RFCs.
- ITU-T:** Telecommunications sector.

Service Access Points (SAPs): Layer (N-1) offers service to Layer N through SAPs using Primitives: Request, Indication, Response, Confirm.

1. OSI 7-Layer Reference Model (ISO 7498) Functional Deep Dive

#	Layer Name	PDU	Primary Core Functions & Protocols	Addressing & Hardware
7	Application	Data	Direct network interface for user software: File transfer (FTP), Web browsing (HTTP/HTTPS), Email (SMTP/POP3/IMAP), Name resolution (DNS), Dynamic config (DHCP).	URL / URI, Email IDs
6	Presentation	Data	Translation: Character encoding (ASCII, EBCDIC). Encryption/Decryption: Cryptographic security (SSL/TLS). Compression: Lossy & lossless payload reduction.	MIME Types
5	Session	Data	Dialog Control: Enforces simplex, half-duplex, or full-duplex session turns. Synchronization & Checkpoints: Inserts recovery checkpoints in large data streams.	Session IDs, Tokens
4	Transport	Segment / Datagram	"Heart of OSI": End-to-end process-to-process delivery. Segmentation & Reassembly, Port addressing, Connection-oriented (TCP) vs Connectionless (UDP), End-to-end flow & error control.	Port Numbers (16-bit: 0 - 65535)
3	Network	Packet	Source-to-destination Host-to-Host delivery across heterogeneous internetworks. Logical IP addressing, Routing algorithms (Dijkstra, Distance Vector), Packetizing & fragmentation.	Logical IP Address (32/128-bit) <i>Routers, L3 Switches</i>
2	Data Link	Frame	Hop-to-hop Node-to-Node frame delivery. Framing (Bit/Byte stuffing), Physical MAC addressing, Flow control, Error control (CRC Trailer), Shared medium access control (CSMA/CD).	Physical MAC Address (48-bit) <i>Bridges, L2 Switches, NIC</i>
1	Physical	Bits	Transmission of raw binary stream (0s and 1s) over physical media. Voltage levels, Bit duration, Line coding (Manchester/NRZ), Transmission mode (Simplex/Duplex), Pinouts (RJ-45).	Signal Voltages, Optical pulses <i>Repeaters, Hubs, Modems</i>

2. Encapsulation & Trailer T2 Anatomy

- **Encapsulation (Top-to-Bottom):** Har layer apna protocol header append karti hai (\$H_7 o H_6 o H_5 o H_4 o H_3 o H_2\$).
- **Unique DLL Trailer (\$T_2\$):** Data link layer akeli layer hai jo Header (\$H_2\$) ke sath **Trailer (\$T_2\$)** bhi append karti hai.
- \$T_2\$ carries **CRC Checksum** hardware bit-error detection ke liye!
- **Decapsulation (Bottom-to-Top):** Receiving host par headers inspect aur strip hote hain.

3. The 3 Distinct Delivery Scopes (AKTU Must-Know)

- **1. Hop-to-Hop (Data Link Layer):** Frame ko ek router/switch se agle adjacent router tak move karta hai using **MAC Addresses**. MAC changes at every hop!
- **2. Host-to-Host (Network Layer):** Packet ko source host computer se destination host tak deliver karta hai using **IP Addresses**. IP stays constant end-to-end!
- **3. Process-to-Process (Transport Layer):** Packet ko computer ke exact running application (e.g. Chrome tab) tak deliver karta hai using **Port Numbers**.

4. Master 10-Point Comparison: OSI Model vs TCP/IP Protocol Suite

Comparison Dimension	OSI Reference Model	TCP/IP Protocol Suite
Origin & Design Goal	ISO standard (1984); Conceptual theoretical reference model.	ARPANET / DoD (1970s); Practical implementation for working network.
Layer Architecture	Strictly 7 Layers .	4 Layers (Network Access, Internet, Transport, Application).
Protocol vs Service Separation	Clear, strict separation (Service, Interface, Protocol).	Loose separation; Layers are built around specific protocols.
Network Layer Service	Supports both Connection-Oriented & Connectionless .	Strictly Connectionless (Internet Protocol - IP datagrams).
Transport Layer Service	Strictly Connection-Oriented .	Supports both Connection-Oriented (TCP) & Connectionless (UDP) .
Session & Presentation Layers	Separate dedicated layers with distinct protocols.	No separate layers; Functions handled inside Application layer.
Protocol Independence	Protocols easily replaceable; Model developed before protocols.	Protocols tightly coupled; Hard to replace IP or TCP.
Global Commercial Dominance	Never gained widespread commercial adoption.	Universal standard powering the entire global Internet!

5. 4 Levels of Network Addressing in TCP/IP

- **Physical (MAC) Address:** 48-bit (6 Bytes). Hexadecimal format: 00:1A:2B:3C:4D:5E. First 24 bits = OUI (Vendor ID); Last 24 bits = NIC Serial. Burned into hardware.
- **Logical (IP) Address:** Universal host identifier across subnets. IPv4 = 32-bit (4 Bytes dotted-decimal 192.168.1.1); IPv6 = 128-bit (16 Bytes hex).
- **Port Address:** 16-bit integer (0 to 65,535). Well-known: 0-1023 (HTTP 80, HTTPS 443, DNS 53, SSH 22). Registered: 1024-49151. Dynamic: 49152-65535.
- **Specific / Application Address:** User-friendly URLs (www.aktu.ac.in) and emails. DNS resolves URL \$ o\$ IP.

Socket Address = IP Address : Port Number (e.g. 192.168.1.10 : 8080)

1. Transmission Impairments & Decibels

- Attenuation:** Loss of energy over distance. Solved by Amplifiers/Repeaters.

$$dB = 10 \log_{10}(P_2 / P_1) \quad | \quad dB = 20 \log_{10}(V_2 / V_1)$$

Power halved $\implies -3 \text{ dB}$. Multi-stage: $\text{dB}_{\text{total}} = \text{dB}_1 + \text{dB}_2$.
- Distortion:** Phase shift in composite signals (frequencies travel at different velocities).
- Noise:** Thermal (kTB $\$$), Induced, Crosstalk (wire coupling), Impulse (lightning spikes).

$$SNR = P_{\text{signal}} / P_{\text{noise}} \quad | \quad SNR_{dB} = 10 \log_{10}(SNR)$$

2. Nyquist vs Shannon Channel Capacity

- Nyquist (Noiseless Ideal Channel):**

$$C = 2 \cdot B \cdot \log_2(M) \text{ bps} \quad | \quad \text{Baud} = 2 \cdot B$$

B = Bandwidth (Hz), M = Discrete signal voltage levels.
- Shannon (Noisy Real-World Channel):**

$$C = B \cdot \log_2(1 + SNR) \text{ bps} \text{ (Absolute Physical Ceiling)}$$

Linear SNR $= 10^{\{\text{SNR}_{dB}\} / 10}$. If $B=3000 \text{ Hz}$, $\text{SNR}_{dB}=31.62 \text{ dB}$ $\implies C \approx 31.6 \text{ kbps}$.
- Synthesis (Find M):** $2B \log_2(M) \leq C_{\text{Shannon}}$ $\implies M \leq \sqrt{1 + \text{SNR}}$.

3. Network Latency Breakdown (4 Delay Components) & Performance Metrics

- Transmission Delay:** $T_t = \frac{L}{B}$ (Packet Length L / Bandwidth B). Independent of distance!
 - Propagation Delay:** $T_p = \frac{d}{v}$ (Distance d / Wave velocity $v \approx 2 \times 10^8 \text{ m/s}$). Independent of packet size!
 - Queuing Delay (T_q):** Time waiting in intermediate router buffer queues.
 - Processing Delay (T_{proc}):** Header inspection, CRC verify, route lookup.
- $$\text{Total Latency } T = T_t + T_p + T_q + T_{\text{proc}} \quad | \quad \text{RTT} \approx 2 \cdot T_p + T_t$$
$$\text{BDP (Bandwidth-Delay Product)} = \text{Bandwidth} \cdot T_p \text{ (Bits in Flight)}$$
- Throughput:** Actual measured delivery rate ($\leq \text{Bandwidth}$).
 - Goodput:** Application payload delivery rate (excludes headers).
 - Jitter:** Delay variance between successive arriving packets.

4. Digital Transmission & Line Coding Schemes (Bit Stream: 0 1 0 0 1 1 1 0)

Scheme	Rule for Bit '0'	Rule for Bit '1'	Self-Sync	DC Component
Polar NRZ-L	High Voltage (+V)	Low Voltage (-V)	Poor on runs	Moderate
Polar NRZ-I	No change (Stay at current level)	Inversion at start of interval	Syncs on 1s only	Moderate
Polar RZ	Negative (-V) returns to 0 V at mid-bit	Positive (+V) returns to 0 V at mid-bit	Good	None
Manchester (802.3)	High-to-Low transition (↓) at mid-bit	Low-to-High transition (↑) at mid-bit	Perfect (Every bit)	Zero (None)
Diff Manchester	Transition at START of bit interval	NO transition at start of bit interval	Perfect (Mid-bit)	Zero (None)
Bipolar AMI	Zero Voltage (0 V)	Alternating +V and -V pulses	Syncs on 1s only	Zero (None)

5. Switching Techniques Compared

- Circuit Switching:** Dedicated physical path. 3 Phases (Setup $\rightarrow$ Data $\rightarrow$ Teardown). Zero queuing delay, but bandwidth wasted during silence. (PSTN Telephone).
- Message Switching:** Entire message stored and forwarded at every node. Huge hard drive buffers required. High delay. (Telegraph).
- Packet Switching:** Small packets (MTU). Statistical multiplexing, concurrent pipelining.
 - Datagram (Connectionless):* Packets routed independently; can arrive out-of-order; robust. (IP).
 - Virtual Circuit (Connection-oriented):* Pre-planned path (SVC/PVC); in-order arrival; uses VCI. (ATM).

6. Network Devices & Domain Separation Matrix

Device	Layer	Collision Domain	Broadcast Domain
Repeater / Hub	L1 Physical	1 (Single shared pool)	1 (Floods all)
Bridge / Switch	L2 Data Link	N (1 per port)	1 (Floods broadcast)
Router	L3 Network	N (1 per port)	N (Blocks broadcast)
Gateway	L4-7 All	Separate	Separate

Golden Rule: Switches break Collision Domains. Routers break BOTH Collision and Broadcast Domains!